INCH-POUND

MIL-C-24643/45C
14 March 1994
SUPERSEDING
MIL-C-24643/45B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2U (INCLUDING VARIATION LS2UA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2U (unarmored) and LS2UA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

	First	- Copper conductor, tin coated, ASTM B 286, size 26-7. Maximum diameter 0.020 inch.
	Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
	Third	- Crosslinked polyethylene insulation. Nominal wall thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Standard identification code shall be applied by method 3.
*	Fourth	- Two conductors cabled together with a maximum lay of 1-1/2 inches to form a pair.
	Fifth	- The specified number of pairs (see table I) cabled together with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward.
	Sixth	- Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
*	Seventh	- Braided shield of tin coated copper. Minimum coverage of 85 percent.

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Construction (non-watertight) (continued)

Construction variant for type LS2U

* Eighth - Cross-linked polyolefin jacket (see table I for thickness).

Construction variant for type LS2UA

* Eighth - Cross-linked polyolefin jacket (see table I for thickness).
* Ninth - Braided metal armor.

* TABLE I. Details.

Military 1/ part no. M24643/45	Size 2/	Number of pairs	Cable jacket thickness (min avg) (inch)	Dia over Shielded core (inch nom)	Overall diameter (inches)			
					TYPE LS2U		TYPE LS2UA	
					min	max	min	max
-01UO	10	10	0.050	0.336	0.450	0.480	0.500	0.530
-02UO	15	15	.050	.393	.530	.560	.580	.610
-03UO	19	19	.050	.423	.550	.580	.600	.630
-04UO	30	30	.050	.545	.670	.700	.720	.750
-05UO	45	45	.050	.670	.830	.870	.880	.920
-06UO	60	60	.065	.750	.920	.960	.970	1.010

1/ Unarmored, overall shield. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS2U and LS2UA.

EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	45.19
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
* Conductor to shield	1000
Insulation resistance - Megohms/1000 feet minimum	
* Conductor to conductor	500*
* Conductor to shield	500
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	135 ± 15

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EXAMINATION AND TESTS (continued):	Requirements
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket	50
Cold bending, cable mandrel diameter 12 times cable diameter	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum	25
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
*Permanence of printing (jacket) - (type LS2U only) cycles, minimum	125
Armor (type LS2UA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
*Surface transfer impedance (type LS2U only)	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60
* Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield Insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet (nominal)

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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